

1. Explain how ML can identify poverty-prone regions using socio-economic data

SDG 1 - No poverty

Machine learning can analyze socio-economic data to identify areas that are more likely to suffer from poverty.

Data used:

- * Household Income
- * Employment rate
- * Education level
- * Population
- * Access to healthcare
- * Housing conditions
- * Access to clean water and electricity

Working:

- Collect socio-economic data from different regions
 - Clean and preprocess the data
 - Select important features such as income, employment & education
 - Train an ML model using historical poverty data
 - The model identifies regions with high poverty risk
 - Governments can use the results to provide targeted financial and social support
- Conclusion: ML identifies poverty prone regions accurately

Q. Design a ML model to identify eligible beneficiaries for government welfare schemes
SDG-1 No poverty

A Machine learning model can be designed to automatically identify people who are eligible for government welfare schemes based on their socio-economic conditions

Input data :

Annual Income

Family size

Employment status

Age

Disability status

Housing condition

Existing government benefits

Steps:

1. Collect beneficiary and eligibility data
2. Clean missing & incorrect values
3. Select relevant features
4. Train a classification algorithm such as decision tree, Random Forest or logistic Regression

5. The model predicts whether a person is eligible or not eligible
6. The result can be verified by government

Conclusion: ML can make welfare distribution faster & more targeted.

3.

Explain how ML can predict crop yield using historical agricultural data

SDG 2 - Zero Hunger

Machine learning can predict expected crop yield by learning patterns from previous agricultural data.

Data used

Rainfall
 Temperature
 Soil Type
 Soil nutrients
 Fertilizer usage
 Irrigation
 Crop type
 Previous crop yield

Working

1. collect historical agricultural data
2. Clean & preprocess the data
3. Select important features
4. Train a regression model such as Linear Regression, Decision Tree Regression or Random Forest Regression
5. Provide current agricultural conditions as input
6. The model predicts the expected crop yield.

Conclusion It helps farmers make better decisions

4. Develop a ML solution for crop disease detection using leaf images
SDG 2 - Zero Hunger

Machine learning can detect crop disease by analyzing photographs of plant leaves

Input: Images of healthy & diseased leaves

Working:

1. Collect images of healthy & diseased leaves.

2. Label the images according to disease type.

3. Resize & preprocess the images

4. Extract useful image feature

5. Train an image-classification model
Such as a CNN (Convolutional Neural Network)

6. Give a new leaf image to trained model

7. The model predicts whether the leaf is healthy or diseased & identifies the disease

Conclusion: ML based crop disease detection helps farmers identify diseases early & protect crops supporting SDG 2 - Zero Hunger

6. Explain how Machine Learning can reduce food wastage in supermarkets.

Machine learning can predict the demand for food products and help supermarkets maintain current stock level.

Data Used:

- Previous sales
- Product type
- Price
- Seasonal demand
- Expiry dates
- Holidays.
- Customer buying patterns
- Current Inventory.

Working:

1. Collect historical sales and inventory data
2. Preprocess the data.
3. Analyze customer purchasing patterns.
4. Train an ML demand prediction model.
5. Predict future demand for each product.
6. Adjust the quantity ordered and stocked,
7. Products nearing expiry can be discounted or distributed.